3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, r and c, representing the number of rows and the number of columns.
- The next r lines contain c space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Sample Test Cases

numpyarr...

```python
import numpy as np

# write your code here...
r, c = map(int, input().split())


elements = []
for _ in range(r):
    row = list(map(int, input().split()))
    elements.extend(row)


arr = np.array(elements).reshape(r, c)

# Output results
print(arr)
print(arr.ndim)
print(arr.shape)
print(arr.size)
```

Average time	Maximum time
0.005 s (4.58 ms)	0.012 s (12.00 ms)

2 out of 2 shown test case(s) passed
10 out of 10 hidden test case(s) passed

Test case 1 7 ms Debug

Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[1 2 3 4]	[[1 2 3 4]
[5 6 7 8]	[5 6 7 8]

Terminal Test cases

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3.2.1. Numpy: Matrix Operations

00:42

The given code takes two 3×3 matrices, $\mathrm{matrix\_a}$, and $\mathrm{matrix\_b}$, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** $(\mathrm{matrix\_a} + \mathrm{matrix\_b})$
2. **Subtraction** $(\mathrm{matrix\_a} - \mathrm{matrix\_b})$
3. **Element-wise Multiplication** $(\mathrm{matrix\_a} * \mathrm{matrix\_b})$
4. **Matrix Multiplication** $(\mathrm{matrix\_a} \cdot \mathrm{matrix\_b})$
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for $\mathrm{matrix\_a}$, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for $\mathrm{matrix\_b}$, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Sample Test Cases +

matrixOp... Submit

```python
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])


# Addition
print("Addition (A + B):")
print(matrix_a+matrix_b)
# Subtraction
print("Subtraction (A - B):")
print(matrix_a-matrix_b)
# Multiplication (element-wise)
print("Element-wise Multiplication (A * B):")
print(matrix_a*matrix_b)
# Matrix multiplication (dot product)
```

Average time 0.007 s 7.25 ms

Maximum time 0.010 s 10.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 10 ms Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
1 1 1	1 1 1

Terminal Test cases

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  - 3.2.7. Student Data Analysis and Operations
- 4. Practical 4
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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

03:49

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases +

stacking.py Submit

```python
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)
print("Horizontal Stack:")
print(np.hstack((arr1, arr2)))



# Perform vertical stacking (vstack)
print("Vertical Stack:")
print(np.vstack((arr1, arr2)))
```

Average time	Maximum time
0.007 s (7.00 ms)	0.008 s (8.00 ms)

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 8 ms Debug

Expected output	Actual output
Enter Array1:	Enter Array1:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Array2:	Enter Array2:
4 5 6	4 5 6

Terminal Test cases

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3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using `numpy` based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

customS...

```python
import numpy as np

# Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())

# Generate the sequence using np.arange()
sequence = np.arange(start, stop, step)
# Print the generated sequence
print(sequence)
```

Average time 0.004 s 4.25 ms

Maximum time 0.006 s 6.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 6 ms Debug

Expected output	Actual output
3	3
10	10
2	2
[3 5 7 9]	[3 5 7 9]

Test case 2 5 ms

Terminal Test cases

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3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bit...

00:21

You are given two arrays `A` and `B`. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. **Arithmetic Operations**:
   - Compute the element-wise sum, difference, and product of the two arrays.
2. **Statistical Operations**:
   - Calculate the mean, median, and standard deviation of array `A`.
3. **Bitwise Operations**:
   - Perform bitwise `AND`, bitwise `OR`, and bitwise `XOR` on the arrays (ex: A_1 `OR` B_1).

Input Format:
- The first line contains space-separated integers representing the elements of array `A`.
- The second line contains space-separated integers representing the elements of array `B`.

Output Format:
- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases +

different... Submit

```python
import numpy as np

def array_operations(A, B):

        # Convert lists to NumPy arrays
    A = np.array(A)
    B = np.array(B)

# Arithmetic operations
    sum_result = A + B
    diff_result = A - B
    prod_result = A * B

# Statistical operations
    mean_A = np.mean(A)
    median_A = np.median(A)
    std_dev_A = np.std(A)

# Bitwise operations
    and_result = A & B
```

Average time	Maximum time
0.005 s 4.67 ms	0.007 s 7.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 7 ms Debug

Expected output	Actual output
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: 6 8 10 12	Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4	Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32	Element-wise Product: 5 12 21 32
Mean of A: 2.5	Mean of A: 2.5

Terminal Test cases

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3.2.5. Numpy: Copying and Viewing Arrays

00:23

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

```
Original array after modifying view: <original_array>
View array: <view_array>
```

- After modifying the copy:

```
Original array after modifying copy: <original_array>
Copy array: <copy_array>
```

Sample Test Cases +

copyAnd... Submit

```python
import numpy as np

inputlist = list(map(int,input().split(" ")))

# Original array
original_array = np.array(inputlist)

# Create a view
view_array = original_array.view()

# Create a copy
copy_array = original_array.copy()


# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)

# Modify the copy
```

Average time	Maximum time
0.003 s 3.00 ms	0.004 s 4.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 4 ms Debug

Expected output	Actual output
10 20 30 40 50 60 70 80	10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]	Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]	View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]	Original array after modifying copy: [99 20 30 40 50 60 70 80]

Terminal Test cases

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3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

00:10

The given code in the editor takes a single array, `array1`, as space-separated integers as input from the user.
Additionally, it takes the following inputs:

- `search_value`: The value to search for in the array.
- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. **Searching**: Find the indices where `search_value` appears in `array1` and print these indices.
2. **Counting**: Count how many times `count_value` appears in `array1` and print the count.
3. **Broadcasting**: Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
4. **Sorting**: Sort `array1` in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing `array1`.
2. An integer `search_value` represents the value to search for in the array.
3. An integer `count_value` represents the value to count in the array.
4. An integer `broadcast_value` represents the value to add to each element of the array.

Output Format:

1. The indices where `search_value` occurs in `array1`.
2. The count of occurrences of `count_value` in `array1`.
3. The array after adding the `broadcast_value` to each element.
4. The sorted array.

Sample Test Cases +

arrayOpe... Submit

```python
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split())))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))

indices = np.where(array1 == search_value)[0]
print(indices)

# 2. Counting occurrences
count = np.count_nonzero(array1 == count_value)
print(count)

# 3. Broadcasting (add value to each element)
broadcast_result = array1 + broadcast_value
print(broadcast_result)
```

Average time	Maximum time
0.005 s	0.008 s
5.50 ms	8.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 8 ms Debug

Expected output	Actual output
1 1 1 2 2 2	1 1 1 2 2 2
Value to search: 1	Value to search: 1
Value to count: 2	Value to count: 2
Value to add: 2	Value to add: 2
[0 1 2]	[0 1 2]
3	3

Terminal Test cases

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3.2.7. Student Data Analysis and Operations

00:13

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.
- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.
- **Find total marks of students**: Compute the total marks for each student across all subjects.
- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2**: Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3**: Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3**: Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2**: Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2**: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1**: Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2**: Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored >=90 in each subject**: Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored >=90**: Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order**: Sort and print the marks of students in Subject 1 in ascending order.

Explorer | Operatio... | Submit | Debugger

```python
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

# All student details
print("All student Details:")
print(" "+str(a))   # FIX 1: correct spacing (no underscore issue)

# Total students
print("Total Students:", a.shape[0])

# Roll numbers
roll = a[:,0]
print("All Student Roll Nos", roll)

# Subject 1
s1 = a[:,1]
print("Subject 1 Marks", s1)

# Subject 2 & 3
```

Average time **0.008 s** 8.00 ms | Maximum time **0.008 s** 8.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 8 ms Debug

Expected output	Actual output
All student Details:	All student Details:
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]

Terminal | Test cases

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